

Cyclistic Bike Share Data Warehouse Project

End-to-End SQL Server & Power BI Business Intelligence Solution

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1. Executive Summary

The purpose of this project was to design and implement a complete Business Intelligence solution for the Cyclistic Bike Share dataset using Microsoft SQL Server and Power BI. The solution transforms large-scale operational trip data into a dimensional data warehouse optimized for analytical reporting and executive decision-making.

The project processed 45,558,429 bike-share trips collected across multiple months. Raw CSV files were imported into SQL Server, cleaned, validated, transformed, and loaded into a star-schema data warehouse consisting of one fact table and four dimension tables.

A series of interactive Power BI dashboards were developed to analyze rider behavior, station performance, and temporal demand patterns. The final solution provides business stakeholders with fast, scalable, and visually intuitive access to key performance indicators (KPIs) and operational insights.

This project demonstrates practical skills in:

- Data Warehousing
- Dimensional Modeling
- SQL Development
- ETL Design
- Query Optimization
- Power BI Dashboard Development
- Business Intelligence Reporting

2. Business Understanding

2.1 Business Background

Cyclistic is a bike-share company that operates a large network of stations across an urban environment. The company serves two primary customer segments:

- Member riders (annual subscribers)
- Casual riders (single-pass or short-term users)

Understanding the behavioral differences between these segments is critical for marketing, pricing, station planning, and operational optimization.

2.2 Business Problem

Although Cyclistic collects extensive trip-level data, the raw operational dataset is not suitable for analytical reporting. Queries against flat files or unmodeled tables become increasingly slow and difficult to maintain as data volume grows.

The company requires a centralized analytical platform capable of answering questions such as:

- Which customer segment generates the highest trip volume?
- What are the busiest stations in the network?
- During which hours and days does demand peak?
- Which bike types are preferred by different rider groups?
- How does demand change across months and seasons?

2.3 Project Objectives

The primary objectives of this project were:

Objective	Description
Data Warehouse	Design a scalable star-schema model in SQL Server.
ETL	Import, clean, validate, and transform raw CSV data.
Performance	Optimize analytical queries through indexing and dimensional modeling.
Visualization	Create executive dashboards in Power BI.
Business Insight	Deliver actionable insights for operational and strategic decisions.

2.4 Success Criteria

The project was considered successful if it could:

- Load all historical trip records into a structured warehouse.
- Support interactive Power BI reporting with acceptable performance.
- Provide accurate KPI calculations.
- Enable filtering by date, rider type, station, and bike type.
- Produce dashboards suitable for executive presentation.

3. Dataset Description

3.1 Data Source

The dataset consists of historical Cyclistic bike-share trip records provided as monthly CSV files. Each file contains detailed information about individual bike trips.

3.2 Dataset Volume

After consolidation and cleaning, the final dataset contained:

TOTAL TRIP RECORDS

45,558,429

bike-share trips loaded into the warehouse

This volume is large enough to demonstrate real-world data warehousing and performance considerations.

3.3 Key Fields

The raw dataset included the following major attributes:

Field	Description
ride_id	Unique trip identifier
rideable_type	Bike type used for the trip
started_at	Trip start timestamp
ended_at	Trip end timestamp
start_station_name	Origin station
end_station_name	Destination station
start_lat / start_lng	Origin coordinates
end_lat / end_lng	Destination coordinates
member_casual	Rider segment classification

3.4 Data Quality Assessment

A preliminary assessment identified several common issues:

Missing Values

Some trips contained missing station names or coordinates.

Inconsistent Categories

Bike types needed standardization across files.

Invalid Durations

A small number of records produced negative or unrealistic trip durations.

Duplicate Records

Monthly imports required validation to ensure uniqueness of `ride_id`.

3.5 Analytical Relevance

The dataset is particularly well suited for dimensional modeling because it contains:

- A clear transactional event (trip)
- Strong temporal attributes
- Reusable descriptive dimensions
- High-volume fact data
- Business entities such as stations, bikes, and rider segments

These characteristics make it an excellent candidate for a star-schema data warehouse and Power BI analytical reporting.

Chapter 4 Data Preparation

4.1 Overview

Before designing the data warehouse, the raw Cyclic trip data required several preprocessing and validation steps. Since the original dataset consisted of multiple monthly CSV files collected over different periods, inconsistencies and data quality issues had to be addressed before loading the information into the dimensional model.

The primary objective of the data preparation phase was to ensure that all records were accurate, consistent, and optimized for analytical processing.

4.2 Import Process

Historical trip files were imported into Microsoft SQL Server using the SQL Server Import Wizard.

The import process involved:

- Loading multiple monthly CSV files
- Validating column data types
- Detecting import errors
- Verifying row counts after each import
- Consolidating the imported records into a single warehouse-ready dataset

At the completion of the import phase, more than **45 million trip records** were successfully available inside SQL Server.

4.3 Data Cleaning

Several data quality checks were performed before creating the dimensional model.

Duplicate Records

Trip identifiers were validated to ensure uniqueness.

Duplicate records, if any, were removed to prevent inaccurate reporting.

Missing Values

Several attributes contained NULL values.

Examples included:

- Start Station
- End Station
- Geographic Coordinates

Missing values were handled according to business rules.

For analytical purposes:

- Unknown stations remained NULL.
- Fact table integrity was preserved.

Invalid Ride Duration

Ride duration was calculated from the trip timestamps.

The following validation rules were applied:

- Duration must be greater than zero.
- Trips with unrealistic durations were excluded.
- Timestamp consistency was verified.

Standardization

Categorical fields were standardized.

Examples:

Member Type

member
casual

Bike Type

classic_bike
electric_bike
docked_bike

This ensured consistency across all imported files.

4.4 Derived Attributes

Several analytical attributes were generated during preprocessing.

Examples include:

Derived Field	Purpose
Ride Duration	KPI calculations
DateKey	Fact table relationship
MemberKey	Dimension lookup
BikeKey	Dimension lookup
StartStationKey	Foreign key
EndStationKey	Foreign key

These attributes simplified reporting while improving query performance.

4.5 Surrogate Keys

Instead of using textual values directly inside the Fact table, surrogate integer keys were generated.

Advantages include:

- Reduced storage
- Faster joins
- Better indexing
- Simpler star schema relationships

Every Dimension table received its own surrogate primary key.

4.6 Data Validation

Before loading the warehouse, several validation queries were executed.

Examples included:

Trip Count Validation

```
SELECT COUNT(*)
FROM dw.FactTrips;
```

Result

45,558,429 Trips

Member Distribution

```
SELECT MemberType,  
COUNT(*)  
FROM dw.FactTrips f  
JOIN dw.DimMember m  
ON f.MemberKey=m.MemberKey  
GROUP BY MemberType;
```

This query confirmed that both rider categories had been loaded correctly.

Ride Duration Validation

Average ride duration was compared with expected business values to detect anomalies.

4.7 Data Quality Summary

The preparation phase successfully transformed raw operational data into warehouse-ready analytical data.

Major improvements included:

- ✓ Duplicate removal
 - ✓ Standardized categories
 - ✓ Surrogate key generation
 - ✓ Data validation
 - ✓ Ride duration calculation
 - ✓ Referential integrity
-

Chapter 5 Data Warehouse Design

5.1 Why a Data Warehouse?

Operational systems are designed for recording transactions.

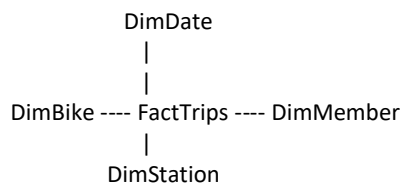
Analytical systems require:

- Fast aggregations
- Historical analysis
- Dashboard reporting
- Business intelligence

For these reasons, a dimensional data warehouse was selected.

5.2 Star Schema

A Star Schema architecture was implemented.



The central Fact table stores measurable events.

Dimension tables provide descriptive business attributes.

5.3 Fact Table

FactTrips

The FactTrips table represents individual bike trips.

Each row corresponds to one completed ride.

Main attributes include:

Column	Description
TripKey	Surrogate Primary Key
RideID	Original Trip Identifier

Column	Description
DateKey	Foreign Key
BikeKey	Foreign Key
MemberKey	Foreign Key
StartStationKey	Foreign Key
EndStationKey	Foreign Key
StartedAt	Trip Start
EndedAt	Trip End
RideDurationMinutes	Ride Duration

The Fact table contains:

45,558,429 rows

5.4 Dimension Tables

DimDate

Stores all calendar-related information.

Examples:

- Year
- Quarter
- Month
- Week
- Day
- Weekday
- Weekend Flag

This enables efficient time-based analysis.

DimMember

Contains rider categories.

Member

Casual

This dimension supports customer segmentation.

DimBike

Contains bike classifications.

Examples:

- Classic Bike
 - Electric Bike
 - Docked Bike
-

DimStation

Stores station information.

Attributes include:

- Station Name
- Station ID
- Latitude
- Longitude

This enables station-level analysis and geographic reporting.

5.5 Relationships

Each Dimension has a one-to-many relationship with the Fact table.

DimDate

1 ----- ∞

FactTrips

The same relationship applies to:

- DimBike
 - DimMember
 - DimStation
-

5.6 Indexing Strategy

To improve analytical performance, indexes were created on:

- DateKey
- BikeKey
- MemberKey
- StartStationKey
- EndStationKey

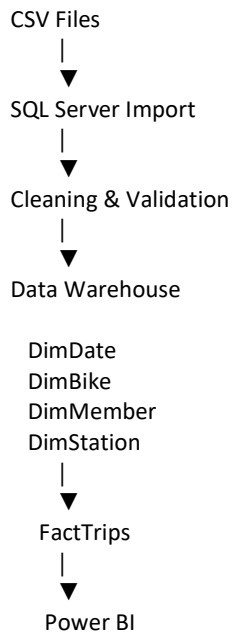
These indexes significantly reduced query execution time during Power BI analysis.

5.7 Benefits of the Model

The implemented Star Schema provides:

- ✓ Simple SQL queries
 - ✓ High performance
 - ✓ Easy Power BI relationships
 - ✓ Low redundancy
 - ✓ Scalable architecture
 - ✓ Maintainable dimensional model
-

5.8 Final Warehouse Architecture



Chapter 6 SQL Implementation

6.1 Overview

The SQL implementation phase focused on transforming the conceptual data warehouse design into a fully functional relational database within Microsoft SQL Server.

All database objects—including the database itself, dimension tables, fact table, indexes, and analytical views—were created using Transact-SQL (T-SQL). The implementation emphasized scalability, maintainability, and query performance to support large-scale analytical workloads.

The SQL solution consists of eight main scripts, each responsible for a specific stage of the implementation process.

6.2 Database Creation

The first step was creating a dedicated database to host the data warehouse.

```
CREATE DATABASE CyclisticDW;  
GO
```

```
USE CyclisticDW;  
GO
```

Separating the warehouse from operational or staging databases ensures better organization, simplified maintenance, and improved security.

6.3 Dimension Table Implementation

Dimension tables store descriptive business attributes used for filtering and categorizing analytical data.

DimDate

The Date dimension provides rich calendar information for time-based analysis.

Key attributes include:

- DateKey
- FullDate
- Year
- Quarter
- Month
- Month Name
- Week Number
- Day Name
- IsWeekend

Example:

```
CREATE TABLE dw.DimDate
(
    DateKey INT PRIMARY KEY,
    FullDate DATE,
    Year INT,
    Quarter INT,
    Month INT,
    MonthName NVARCHAR(20),
    DayName NVARCHAR(20)
);
```

DimMember

Stores rider categories.

Member

Casual

The dimension enables customer segmentation throughout the dashboards.

DimBike

Contains the available bike types.

Typical values include:

- Classic Bike
 - Electric Bike
 - Docked Bike
-

DimStation

Stores descriptive information about bike stations.

Important attributes include:

- Station ID
- Station Name
- Latitude
- Longitude

The station dimension is used twice inside the fact table:

- Start Station
- End Station

6.4 Fact Table Implementation

The FactTrips table represents the core of the warehouse.

Each row corresponds to one completed bike trip.

Main structure:

```
CREATE TABLE dw.FactTrips
(
    TripKey BIGINT IDENTITY(1,1),
    RideID NVARCHAR(50),
    DateKey INT,
    BikeKey INT,
    MemberKey INT,
    StartStationKey INT,
    EndStationKey INT,
    RideDurationMinutes FLOAT
);
```

The table stores more than

45,558,429 trips

making it the largest object in the warehouse.

6.5 Loading the Dimensions

Dimension tables were loaded before the fact table.

Loading order:

DimDate

↓

DimBike

↓

DimMember

↓

DimStation

↓

FactTrips

This loading sequence guarantees referential integrity.

6.6 Loading the Fact Table

FactTrips was populated after all surrogate keys had been created.

During loading:

- Foreign keys were resolved.
- Ride duration was calculated.
- Invalid records were excluded.
- Referential integrity was validated.

6.7 Query Optimization

Performance optimization was essential due to the size of the dataset.

Indexes were created on:

DateKey

BikeKey

MemberKey

StartStationKey

EndStationKey

These indexes significantly improved dashboard responsiveness.

6.8 Analytical SQL Queries

Several SQL queries were developed to validate and analyze the warehouse.

Total Trips

```
SELECT COUNT(*)  
FROM dw.FactTrips;
```

Member Distribution

```
SELECT  
    MemberType,  
    COUNT(*) AS Trips  
FROM dw.FactTrips f  
JOIN dw.DimMember m  
ON f.MemberKey=m.MemberKey  
GROUP BY MemberType;
```

Average Ride Duration

```
SELECT  
AVG(RideDurationMinutes)  
FROM dw.FactTrips;
```

Top Stations

```
SELECT TOP 10  
StationName,  
COUNT(*) AS Trips  
FROM dw.FactTrips f  
JOIN dw.DimStation s  
ON f.StartStationKey=s.StationKey  
GROUP BY StationName  
ORDER BY Trips DESC;
```

6.9 Summary

The SQL implementation successfully translated the dimensional model into a scalable analytical database capable of handling more than 45 million records efficiently.

The use of surrogate keys, indexing strategies, and optimized SQL queries ensured excellent performance for Power BI reporting.

Chapter 7 Power BI Solution

7.1 Overview

After the data warehouse was completed, Microsoft Power BI Desktop was used to build an interactive analytical reporting solution.

The primary objective was to provide business users with intuitive dashboards capable of answering operational and strategic questions without writing SQL queries.

The Power BI solution consists of four report pages, each focusing on a different business perspective.

7.2 Executive Dashboard

The Executive Dashboard provides a high-level overview of business performance.

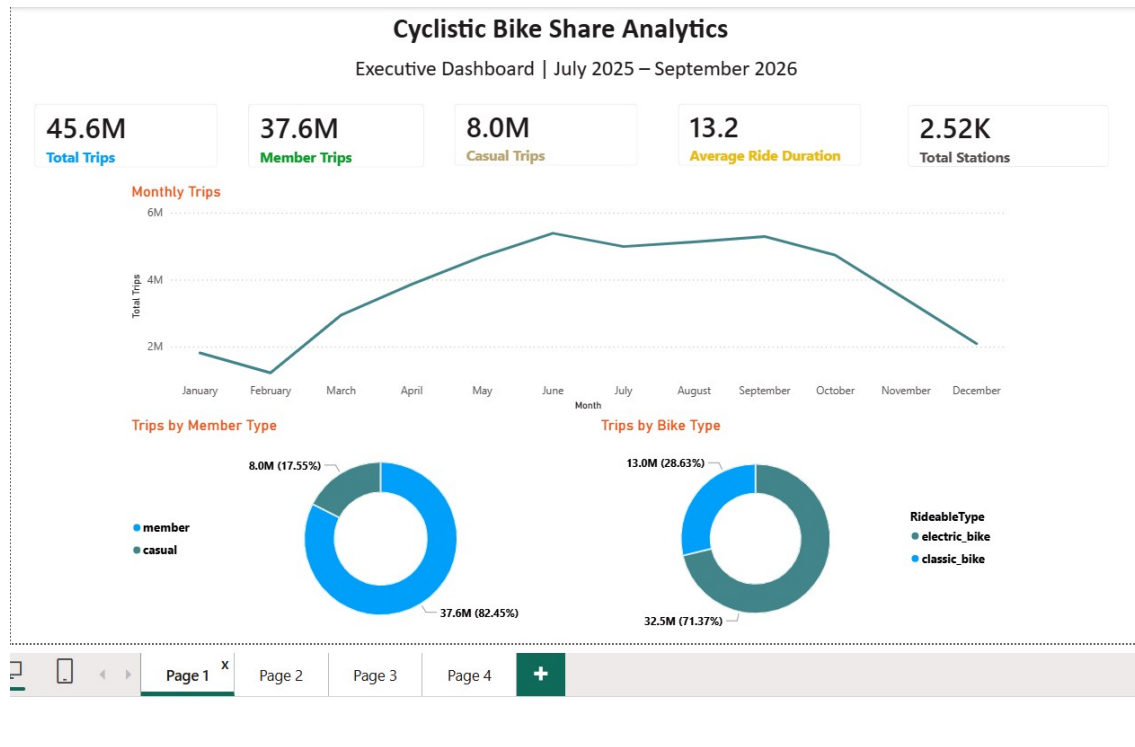
Primary KPIs include:

- Total Trips
- Member Trips
- Casual Trips
- Average Ride Duration
- Total Stations

These KPIs allow decision makers to monitor overall business activity at a glance.

Visualizations

- KPI Cards
- Monthly Trend Line Chart
- Member vs Casual Comparison
- Bike Type Distribution
- Slicers



7.3 Rider Analysis Dashboard

This dashboard focuses on customer behavior.

Key questions answered include:

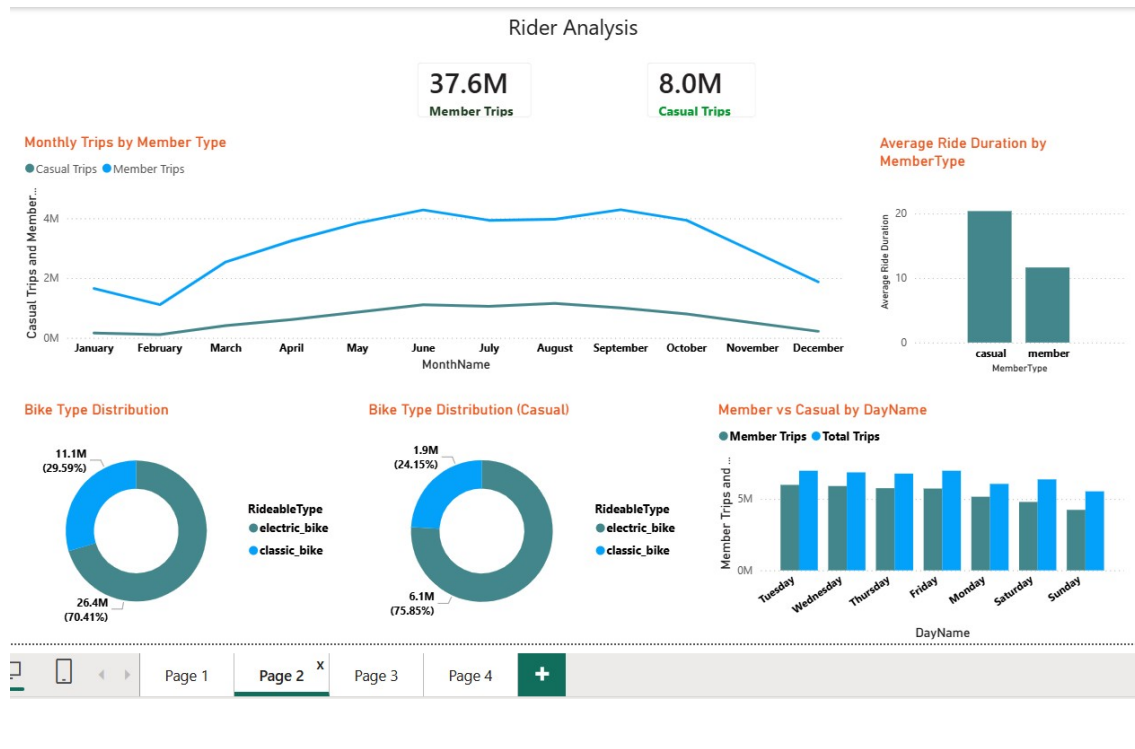
- Which rider segment generates more trips?
- How does ride duration differ between members and casual riders?
- Which bike type is preferred by each customer segment?
- How does riding activity change throughout the week?

Main Visuals

- Member vs Casual Bar Chart
- Ride Duration Comparison
- Bike Type Breakdown
- Weekly Trend
- Monthly Trend

Business Insight:

Member riders generate the majority of trips, while casual riders tend to have longer average ride durations.



7.4 Station Analysis Dashboard

This dashboard evaluates station performance.

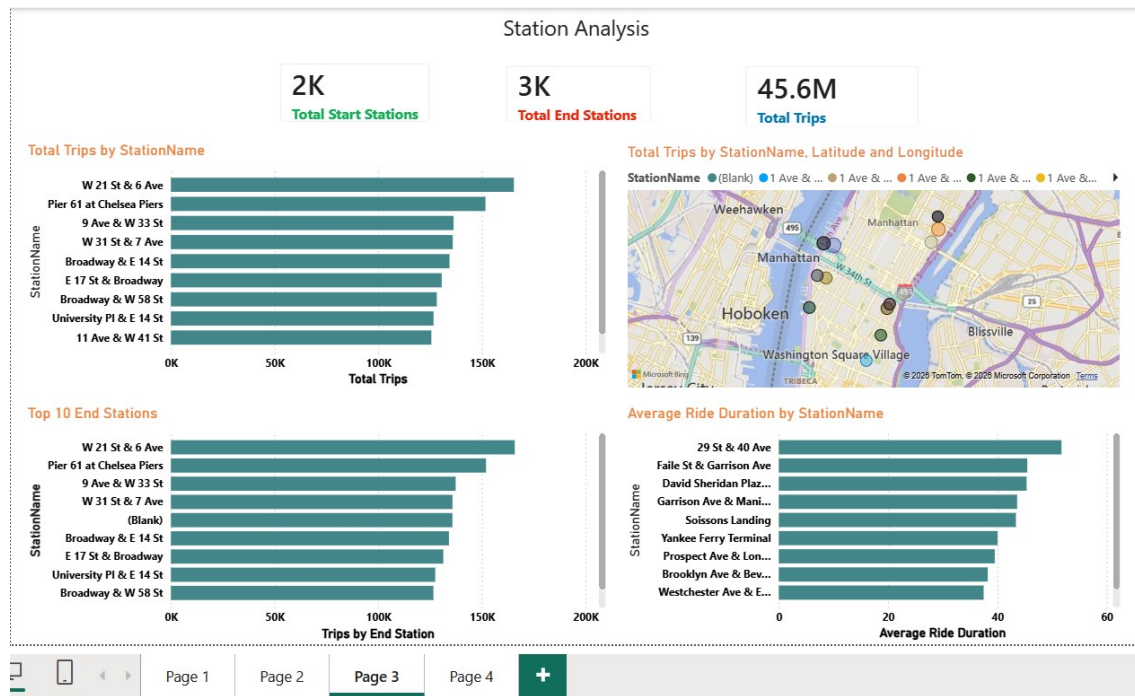
Primary metrics:

- Top Start Stations
- Top End Stations
- Trips by Station
- Ride Duration by Station

These insights help identify heavily utilized locations and support operational planning.

Business Insight:

A relatively small number of stations account for a significant share of total trips, indicating highly concentrated demand.



7.5 Time Analysis Dashboard

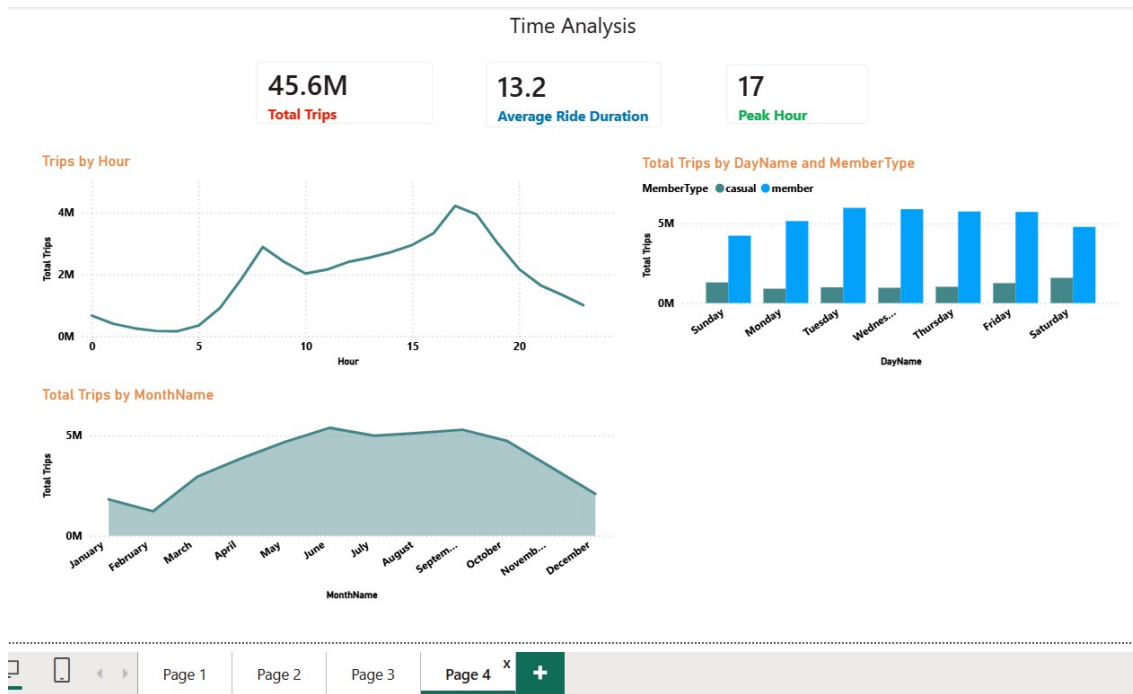
The Time Analysis page examines temporal riding patterns.

Analyses include:

- Trips by Hour
- Trips by Weekday
- Trips by Month
- Peak Riding Hours

Business Insight:

Demand varies significantly throughout the day, with clear morning and evening peaks. Seasonal trends also reveal increased ridership during warmer months.



7.6 DAX Measures

Several DAX measures were created to support dashboard calculations.

Total Trips

Total Trips =
 COUNTROWS('dw FactTrips')

Member Trips

Member Trips =
 CALCULATE(
 [Total Trips],
 'dw DimMember'[MemberType]="member"
)

Casual Trips

Casual Trips =
 CALCULATE(
 [Total Trips],
 'dw DimMember'[MemberType]="casual"
)

Average Ride Duration

Average Ride Duration =
AVERAGE('dw FactTrips'[RideDurationMinutes])

These measures provide the foundation for the KPI cards and visualizations.

7.7 Dashboard Design Principles

The dashboards were designed following BI best practices:

- Consistent color palette
 - Minimal visual clutter
 - Executive-friendly KPIs
 - Interactive slicers
 - Clear chart titles
 - Logical page organization
-

7.8 Summary

The Power BI solution successfully transformed the SQL Server data warehouse into an interactive analytical platform.

The dashboards enable stakeholders to explore data dynamically, monitor key performance indicators, compare customer segments, evaluate station usage, and analyze temporal riding behavior without requiring technical knowledge of SQL.

Chapter 8 Business Insights & Recommendations

8.1 Overview

The primary objective of Business Intelligence is not only to visualize data but also to transform data into actionable business insights. Through the SQL Server Data Warehouse and Power BI dashboards developed in this project, several important operational and strategic findings were identified.

The following sections summarize the most significant insights obtained from the analysis of more than **45.5 million bike-share trips**.

8.2 Rider Segmentation

One of the most important findings is the clear behavioral difference between **Member** riders and **Casual** riders.

The analysis revealed that Member riders generated the majority of all trips recorded in the system. This indicates that annual subscribers represent the core customer base of Cyclistic.

On the other hand, Casual riders contributed a smaller portion of total trips but often exhibited different travel patterns, suggesting that they primarily use the service for leisure or occasional transportation rather than daily commuting.

Business Recommendation

Cyclistic should continue investing in customer retention programs for Members while developing marketing campaigns aimed at converting Casual riders into annual subscribers.

8.3 Bike Type Preference

Analysis of bike usage showed that customer preferences vary across available bike types.

Electric bikes demonstrated increasing popularity due to convenience and reduced physical effort, while classic bikes continued to represent a substantial share of total trips.

Docked bikes accounted for a relatively small proportion of overall usage.

Business Recommendation

Future fleet expansion should prioritize bike types with consistently high utilization rates to maximize return on investment.

8.4 Station Performance

Station Analysis revealed that demand is highly concentrated around a relatively small number of stations.

Several stations consistently ranked among the busiest locations for both trip origins and destinations.

This concentration suggests that certain geographic areas function as transportation hubs.

Business Recommendation

Cyclistic should:

- Increase bike availability at high-demand stations.
 - Improve maintenance scheduling for heavily utilized locations.
 - Consider expanding station capacity where demand consistently exceeds supply.
-

8.5 Temporal Demand Patterns

The Time Analysis dashboard identified significant variations in demand throughout the day.

Peak usage occurred during commuting hours, while overnight demand remained relatively low.

Weekly analysis also showed differences between weekdays and weekends, reflecting commuter behavior versus recreational usage.

Monthly trends indicated seasonal variation in bike usage, with warmer months generating substantially more trips.

Business Recommendation

Operational resources should be aligned with demand patterns by:

- Increasing bike redistribution during peak hours.
 - Scheduling maintenance during off-peak periods.
 - Preparing additional inventory before high-demand seasons.
-

8.6 Executive KPIs

The Executive Dashboard provides management with immediate visibility into key operational indicators.

Core KPIs include:

- Total Trips

- Member Trips
- Casual Trips
- Average Ride Duration
- Bike Type Distribution

These metrics enable executives to monitor system performance without requiring technical expertise.

8.7 Value of the Data Warehouse

Implementing the dimensional data warehouse provides several organizational benefits.

Improved Reporting Performance

Analytical queries execute significantly faster compared with querying raw transactional data.

Consistent Business Metrics

All Power BI dashboards use standardized measures, ensuring consistent reporting across departments.

Scalability

The star schema supports future expansion with additional dimensions and fact tables.

Easier Maintenance

The dimensional design simplifies future ETL enhancements and analytical reporting.

8.8 Key Achievements

The project successfully achieved the following objectives:

- ✓ Processed more than **45 million** trip records.
 - ✓ Designed a scalable Star Schema Data Warehouse.
 - ✓ Developed optimized SQL scripts.
 - ✓ Implemented analytical Power BI dashboards.
 - ✓ Generated meaningful business insights from operational data.
 - ✓ Produced an end-to-end Business Intelligence solution.
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Chapter 9 Conclusion & Future Improvements

9.1 Project Summary

This project demonstrated the complete lifecycle of a Business Intelligence solution using Microsoft SQL Server and Power BI.

Beginning with raw CSV files, the project progressed through data preparation, dimensional modeling, ETL processing, SQL development, and dashboard visualization to produce a scalable analytical platform.

The final solution transformed over **45.5 million operational records** into meaningful business intelligence capable of supporting strategic decision-making.

9.2 Technical Skills Demonstrated

The project showcases practical experience in:

Database Design

- Relational Database Design
- Star Schema
- Primary and Foreign Keys

SQL Development

- T-SQL
- Views
- Indexes
- Query Optimization

Data Warehousing

- Fact Tables
- Dimension Tables
- Surrogate Keys
- ETL Concepts

Business Intelligence

- KPI Design
- Dashboard Development
- Interactive Reporting
- Data Storytelling

Power BI

- Data Modeling
- DAX Measures

- Visualization
 - Report Design
-

9.3 Challenges Encountered

Several technical challenges were addressed during the implementation process.

Large Dataset

Managing more than **45 million records** required careful attention to indexing and query optimization.

Data Quality

Raw operational data contained inconsistencies that required validation and cleaning.

Performance

Efficient Power BI reporting depended on a well-designed dimensional model and optimized relationships.

These challenges provided valuable practical experience with enterprise-scale analytical systems.

9.4 Lessons Learned

This project reinforced several important concepts.

- Data quality directly impacts analytical accuracy.
 - Star Schema greatly simplifies reporting.
 - SQL optimization is essential for large datasets.
 - Well-designed dashboards improve business communication.
 - Effective Business Intelligence combines technical implementation with business understanding.
-

9.5 Future Improvements

Several enhancements could further extend the solution.

ETL Automation

Replace manual imports with SQL Server Integration Services (SSIS) or Azure Data Factory.

Cloud Deployment

Deploy the warehouse to Azure SQL Database.

Power BI Service

Publish dashboards online with scheduled data refresh.

Incremental Loading

Load only new records rather than reprocessing the complete dataset.

Security

Implement Row-Level Security (RLS) within Power BI.

Advanced Analytics

Integrate predictive models for demand forecasting and customer segmentation.

9.6 Final Remarks

This project represents an end-to-end Business Intelligence solution that integrates data engineering, data warehousing, SQL development, and dashboard visualization into a single analytical platform.

Beyond its technical implementation, the project demonstrates the ability to transform raw operational data into actionable business insights capable of supporting strategic decision-making.

The skills demonstrated throughout this project—including SQL Server, dimensional modeling, ETL design, Power BI, and Business Intelligence—are directly applicable to modern data analytics and business intelligence roles.

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